The Penrose transform over F4: a complete computational reference

The finite field F4 supports a rich but structurally constrained version of the Penrose transform. The 85-point twistor space $P^3(F4)$ admits exactly three combinatorial line bundles O(0), O(1), O(2) — unlike the single bundle over F2 or the infinitely many over $\mathbb{C}$. The Hermitian form takes values in F2 (not F4), producing a bipartition into 45 null and 40 non-null points rather than the tripartition one might expect. This bipartition eliminates the PT^+/PT^- distinction, structurally resolving the googly problem for the region structure while leaving a residual bundle-level googly problem for certain helicities. No prior literature exists on this construction; this appears to be genuinely novel mathematics.

The incidence geometry of $P^3(F4)$ in full detail

$P^3(F4)$ consists of 1-dimensional subspaces of $F4^4$. Its parameters follow from Gaussian binomial coefficients (upc) with q = 4:

Structure	Formula	Count
Points	$(q^4-1)/(q-1)$	**85**
Lines	$(q^4-1)(q^3-1)/((q^2-1)(q-1))$	**357**
Planes	$(q^4-1)/(q-1)$	**85**
Points per line	$q+1$	**5**
Lines through each point	q^2+q+1	**21**
Points per plane	q^2+q+1	**21**
Lines per plane	q^2+q+1	**21**
Planes through each line	$q+1$	**5**
Planes through each point	q^2+q+1	**21**

The point-line incidence forms a **2-(85, 5, 1) design**: any two points determine a unique line. (Wikipedia) Total incidences: 85 × 21 = 357 × 5 = **1,785**. Each plane is a copy of PG(2,4) with 21 points and 21 lines. The structure is self-dual — the dual space (points ↔ planes) has identical parameters.

Three line bundles, not infinitely many

The distinction between algebraic and combinatorial line bundles is critical for computation over F4.

In algebraic geometry, the Picard group $\mathrm{Pic}(P^3_k) \cong \mathbb{Z}$ for any field k, (Wikipedia) so O(n) for distinct integers n are always distinct as abstract line bundles. (Wikipedia) The global sections $H^0(P^3, O(n))$ consist of homogeneous polynomials of degree n, with $\dim H^0 = C(n+3, 3)$ for $n \geq 0$ and 0 for $n < 0$. (Wordpress +2) Specifically: $\dim H^0(P^3(F4), O(0)) = \mathbf{1}$, $\dim H^0(O(1)) = \mathbf{4}$, $\dim H^0(O(2)) = \mathbf{10}$, $\dim H^0(O(3)) = \mathbf{20}$.

In the combinatorial framework needed for finite-field computation, a "section of O(n)" is a function $f: F4^4 \setminus \{0\} \to F4$ satisfying $f(\lambda v) = \lambda^n f(v)$ for all $\lambda \in F4^*$. Since **F4* is cyclic of order 3**, we have $\lambda^3 = 1$, so $\lambda^n = \lambda$^(n mod 3). The homogeneity condition depends only on n mod 3, giving exactly **three distinct combinatorial bundles**:

- **O(0)**: $f(\lambda v) = f(v)$ — functions constant on F4*-orbits, i.e., functions on $P^3(F4)$
- **O(1)**: $f(\lambda v) = \lambda f(v)$ — "twisted" functions
- **O(2)**: $f(\lambda v) = \lambda^2 f(v)$ — "doubly twisted" functions

Each combinatorial section space has dimension **85** over F4 (one free value per projective point, since the homogeneity condition merely propagates f(v) to the other two points λv, $\lambda^2 v$ in the orbit). The polynomial sections embed into combinatorial sections via evaluation; for small n this embedding is injective but far from surjective (e.g., 4-dimensional polynomial O(1) sections inside 85-dimensional combinatorial O(1) sections).

Cohomological vanishing holds universally: $H^i(P^3, O(n)) = 0$ for $0 < i < 3$ over any field, including F4. (Stanford University) (columbia) This is a theorem of Serre applicable to the scheme $P^3_\{F4\}$ computed via the standard Čech complex on the affine cover $\{D_+(T_i)\}$. (Wikipedia) (columbia) For the top cohomology, Serre duality gives $H^3(P^3, O(n)) \cong H^0(P^3, O(-4-n))\vee$, which is nonzero only for $n \leq -4$. (Stanford University)

The Hermitian form forces a bipartition, not a tripartition

This is the most important structural clarification. $F4 = \{0, 1, \omega, \omega^2\}$ where $\omega^2 + \omega + 1 = 0$. The **Frobenius automorphism** $\sigma: x \mapsto x^2$ acts as "conjugation," swapping $\omega \leftrightarrow \omega^2$. Its fixed field is **F2 = {0, 1}**.

A Hermitian form h on $F4^4$ satisfies $h(u,v) = \sigma(h(v,u))$, which forces $\mathbf{h(v,v) \in F2}$ for all v. The diagonal form $\Sigma(Z) = Z_0^3 + Z_1^3 + Z_2^3 + Z_3^3$ evaluates to Σ = (number of nonzero coordinates) mod 2, since each Z_i^3 equals 0 or 1. The twistor-theory form $\Sigma = Z_0Z_2^2 + Z_2Z_0^2 + Z_1Z_3^2 + Z_3Z_1^2 = \mathrm{Tr}(Z_0Z_2^2) + \mathrm{Tr}(Z_1Z_3^2)$ (where $\mathrm{Tr}: F4 \to F2$ is $x + x^2$) also takes values in F2.

All non-degenerate Hermitian forms on $F4^4$ are equivalent, so the partition is canonical:

- **PN** (null twistors, $\Sigma = 0$): **45 points** — the Hermitian variety H(3,4)
- **PT** (non-null twistors, $\Sigma = 1$): **40 points**

There is no tripartition. A tripartition would require the form to take ≥ 3 distinct values, but the Hermitian form over $F_\{q^2\}$ with conjugation $x \mapsto xq$ always lands in F_q. Here F_q = F2 has only two elements. To achieve a genuine tripartition, one would need to work over $F_\{q^2\}$ with $q \geq 3$, the smallest case being **F9 = F_{3²}** where the form takes values in F3 = {0, 1, 2}, allowing distinct PT^+ ($\Sigma = 1$) and PT^- ($\Sigma = 2$) regions.

The verification by direct counting confirms 45 + 40 = 85:

Weight (nonzero coords)	Vectors in $F4^4$	Σ value	Projective points
1	$C(4,1)\cdot 3^1 = 12$	1	4
2	$C(4,2)\cdot 3^2 = 54$	0	18
3	$C(4,3)\cdot 3^3 = 108$	1	36
4	$C(4,4)\cdot 3^4 = 81$	0	27
Total	**255**		**85**

PN = 18 + 27 = **45** points. PT = 4 + 36 = **40** points. The symmetry group **PSU(4,2)** (order **25,920** = $2^6\cdot 3^4\cdot 5$) (qmul) acts transitively on each set. Notably, PSU(4,2) ≅ PSp(4,3) — an exceptional isomorphism.

The Klein quadric and the line tripartition

Under the Klein correspondence, lines of PG(3,4) biject with points of the hyperbolic quadric **Q^+(5,4)** in PG(5,4) via Plücker coordinates. (ScienceDirect) The count:

$|Q^+(5,4)| = (q^3-1)(q^2+1)/(q-1) = 63 \times 17 / 3 =$ **357 points** = lines of PG(3,4). ✓

$Q^+(5,4)$ has **1,785 lines** (corresponding to pencils of lines in PG(3,4)), with **25 lines through each point** and two families of **85 maximal planes** (Latin and Greek), (arXiv) corresponding to points and planes of PG(3,4) respectively.

The Hermitian structure on $P^3(F4)$ induces a **tripartition of lines** (not to be confused with the point bipartition):

Line type	$|\ell \cap H(3,4)|$	Count	Description
Generator	5 (all)	**27**	Totally isotropic, entirely in PN
Tangent	1	**90**	Touches H(3,4) at one point
Secant	3	**240**	Meets H(3,4) in a Baer subline

Verification: 27 + 90 + 240 = 357. ✓ Cross-check via incidence: 45 × 3/5 = 27 generators; 45 × 2/1 = 90 tangent lines; 45 × 16/3 = 240 secant lines.

A critical obstruction: every line of PG(3,4) meets H(3,4) in at least 1 point. No line lies entirely in PT. This means there are **zero "purely non-null" spacetime points** in the compactified picture — every line in twistor space intersects the null region. The 27 generators correspond to purely null spacetime points; the remaining 330 lines (90 tangent + 240 secant) correspond to "mixed" points. H(3,4) equipped with its generators forms a **generalized quadrangle GQ(4,2)** with parameters s = 4, t = 2: each of 45 points lies on 3 generators, each generator contains 5 points.

The googly problem partially persists over F4

With three combinatorial bundles, the helicity-to-bundle map $h \mapsto n = -2h - 2 \bmod 3$ produces:

Helicity h	n = −2h−2 mod 3	Bundle	Googly n′ = 2h−2 mod 3	Same?
0	1	O(1)	1	✅ Same
±1/2	0	O(0)	2	❌ Different
±1	2	O(2)	0	❌ Different
±3/2	1	O(1)	1	✅ Same
±2	0	O(0)	2	❌ Different
±5/2	2	O(2)	0	❌ Different
±3	1	O(1)	1	✅ Same

The pattern: opposite helicities ±h share the same bundle precisely when **$2h \equiv 0 \bmod 3$**, i.e., $h \in \{0, \pm 3/2, \pm 3, \pm 9/2, ...\}$. The condition is $4h \equiv 0 \bmod 3$, equivalently $s = 2h \equiv 0 \bmod 3$ (since $\gcd(4,3) = 1$).

The googly problem over F4 has **two layers**:

1. **Region structure** (FULLY RESOLVED): Over $\mathbb{C}$, positive and negative helicity fields live on different regions PT^+ and PT^-. Over F4, since $PT^+ = PT^- = PT$ (no sign in F2), **both helicities live on the same region**. This aspect is completely resolved, identically to F2.
2. **Bundle structure** (PARTIALLY RESOLVED): Over $\mathbb{C}$, h and −h always require different bundles $O(-2h-2) \neq O(2h-2)$. Over F2, there is only O(0), so all helicities share a single bundle (fully resolved). Over F4, the three bundles create **period-3 coincidences**: the googly problem vanishes for $h = 0, \pm 3/2, \pm 3$ but persists for $h = \pm 1/2, \pm 1, \pm 2$. This intermediate behavior — between F2's full resolution and $\mathbb{C}$'s complete googly problem — is the distinctive feature of F4.

The bundle degeneracies also create **helicity collisions**: O(0) serves $\{h = 1/2, 2, 7/2, ...\}$, O(1) serves $\{h = 0, 3/2, 3, ...\}$, and O(2) serves $\{h = 1, 5/2, 4, ...\}$. Distinct physical spins share the same mathematical description.

Spacetime over F4 and the matrix space

The "affine" spacetime consists of 2×2 matrices over F4, parametrizing 2-planes in $F4^4$ that are graphs of linear maps $F4^2 \to F4^2$.

Quantity	Formula	Value
All 2×2 matrices	4^4	**256**
Invertible (det ≠ 0)	$\lvert GL(2,F4)\rvert = (4^2-1)(4^2-4)$	**180**
Singular (det = 0) — null cone	256 − 180	**76**
Nonzero singular (rank 1)	$(4^2-1)^2/(4-1)$	**75**
Zero matrix		**1**
Points at infinity	357 − 256	**101**

GL(2,F4) has order **180** = 4 · 45 = $2^2 \cdot 3^2 \cdot 5$. Its structure: GL(2,4)/Z(GL(2,4)) = PGL(2,4) ≅ A_5 (the alternating group on 5 letters, order 60), with center Z ≅ F4* of order 3. The full group |GL(2,4)| = 3 × 60 = 180.

The 76 singular matrices decompose as: 1 zero matrix plus 75 rank-1 matrices. Each rank-1 matrix is a product uv^T for u, v ∈ $F4^2 \setminus \{0\}$, with the identification (u, v) ~ (λu, λ^{-1}v) for λ ∈ F4*, giving 15 × 15/3 = 75.

The 180 invertible matrices partition by determinant value: det ∈ F4* = {1, ω, ω^2}, and by symmetry each determinant value accounts for 180/3 = **60 matrices** (these are the cosets of SL(2,4), which has order |SL(2,4)| = |GL(2,4)|/(q−1) = 60).

A fundamental cohomological obstruction

The most important computational fact for the Penrose transform is this: **the complement $P^3 \setminus H(3,4)$ is an affine scheme** (it equals $D_+(f)$ where f is the degree-3 polynomial defining H(3,4)), ScienceDirect so

H^i($P^3 \setminus H(3,4)$, O(n)) = 0 for all i > 0 and all n. Stanford University

In the algebraic-geometric (scheme-theoretic) sense, the Penrose transform's input space H^1(PT, O(n)) is trivially zero. This is not a peculiarity of F4 — it holds over any field because removing a hypersurface from projective space yields an affine variety.

This means the **classical cohomological construction of the Penrose transform does not directly produce nontrivial fields over F4**. The transform must be reformulated. Three viable approaches exist:

- **Combinatorial/linear-algebraic**: Work entirely with the 85-dimensional spaces of combinatorial sections and the incidence matrix. Define the "transform matrix" T as a matrix over F4 indexed by (line, point) pairs, with entries weighted by the line bundle character. Then H^1 in the combinatorial sense is computed as ker(∂_1)/im(∂_0) of an appropriate Čech-type complex on the finite point set. This avoids scheme-theoretic cohomology entirely.

- **Relative cohomology**: Use the local cohomology sequence (Wikipedia) $0 \to H^0_H(P^3, O(n)) \to H^0(P^3, O(n)) \to H^0(PT, O(n)) \to H^1_H(P^3, O(n)) \to 0$ to extract information from H^1_H, which is nonzero and computable via the residue sequence for the cubic H(3,4).
- **Restriction to the Hermitian surface**: Since $\omega_H = O_H(-1)$ by adjunction ($\omega_{P^3} = O(-4)$, so $\omega_H = O_H(3-4) = O_H(-1)$), the surface H(3,4) itself carries nontrivial sheaf cohomology, and a modified transform could use cohomology groups of restricted bundles on H.

For building the explicit transform matrix in the combinatorial framework, the key data structures are: the **85 × 357 point-line incidence matrix** I over F4 (or its restriction to the 40 × 330 submatrix for PT), together with the character weights $\chi_n(\lambda) = \lambda^n$ for the three bundles. The weighted incidence matrix for O(n) requires choosing explicit coordinates for each projective point and each line, then incorporating the λ^n factors from the homogeneity.

Conclusion

The Penrose transform over F4 occupies a precise intermediate position between the fully collapsed F2 case (1 bundle, full googly resolution, 5 twistor points) and the classical $\mathbb{C}$ case (infinitely many bundles, full googly problem, continuous twistor space). The three key numbers governing computation are **85** (twistor points), **357** (spacetime points/lines), and **3** (distinct combinatorial bundles). The Hermitian bipartition into 45 + 40 points eliminates the PT^+/PT^- distinction, collapsing one layer of the googly problem, while the mod-3 bundle structure creates a second layer where the googly problem selectively vanishes for spins h with $2h \equiv 0 \bmod 3$. The main obstruction to a direct cohomological construction is that $P^3 \setminus H(3,4)$ is affine, (ScienceDirect) forcing the transform into a combinatorial or local-cohomological framework. No prior literature addresses this construction; the finite geometry ingredients (H(3,4) as GQ(4,2), $Q^+(5,4)$ via Klein correspondence, PSU(4,2) of order 25,920) (Qmul) are all well-established, but their assembly into a Penrose transform is new.